

The Φ_{MDL} Field: Quantum Structure, Born Rule, and Continuum Completion of the Chiral Minkowski CA

Nova Spivack

Independent Researcher

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Abstract

We study the $\mathbb{Z}_7$ -symmetric Klein–Gordon field Φ_{MDL} as the continuum counterpart of the Three-Layer Chiral Minkowski CA (CMCA, Paper P41). Four principal results follow. (1) *Poincaré invariance*: Φ_{MDL} satisfies exact Lorentz invariance $\omega^2 = k^2 + m^2$, machine-certified via `poincare_invariance_of_kg` in `LorentzInvariance.lean` ($\mathbf{A}_{\text{Lean}}$, zero sorry). (2) *Born rule from field amplitudes*: the position-space probability density $P(x) = |\partial_x \Phi|^2 / \int |\partial_x \Phi|^2$ normalizes exactly ($\mathbf{A}/\mathbf{D}$); extended to overlapping multi-kink configurations via the canonical dual-frame inner product (`born_rule_from_dual_frame_master` in `DualFrameBornRule.lean`, $\mathbf{A}_{\text{Lean}}$, zero sorry). (3) *Physical quantum state*: the pre-measurement Φ_{MDL} state is the Hamiltonian thermal ensemble $P(k) = e^{-M_k/T}/Z_T$ restricted to PSC-admissible $\mathbb{Z}_7$ kink sectors $\{0, 2, 3, 4, 6\}$ (`phimdl_thermal_state_master` in `PhiMDLThermalState.lean`, $\mathbf{A}_{\text{Lean}}$ conditional, zero custom physics axioms); the physical state is vacuum-dominant at observed cosmic temperatures and approaches the uniform PSC-admissible distribution at early-universe temperatures. (4) *3+1D extension*: Φ_{MDL} kinks become domain walls with tension $\sigma = 290.10 \text{ MeV GeV}^{-2} = 7450 \text{ MeV fm}^{-2}$; $\mathbb{Z}_7$ superselection is preserved; the Born rule $P(k) = |c_k|^2$ is unchanged; and SR time dilation is exact. When two domain walls intersect in $3 + 1$ dimensions, they form a junction line with exact dimensionless tension $\lambda = -16/49$, corresponding to -1654.77 MeV fm (attractive; ratio $|\lambda/\sigma| = 2/m_\varphi$, pure geometry); certified in `WindingCoinDecoupling.lean` ($\mathbf{A}_{\text{Lean}}$, zero sorry). The continuum limit $\text{CMCA} \rightarrow \Phi_{\text{MDL}}$ is machine-certified in `CMCAContinuumLimit.lean`: algebraic content is M -independent, the Nyquist residual $\varepsilon_0(M) = \pi^2/(3M^2) \rightarrow 0$ as $M \rightarrow \infty$, and Φ_{MDL} is the unique terminal Level-1 Generative Triple Evolution (GTE) object. Two further Lean-certified results complete the picture: `no_finite_ca_exact_lorentz_replica` proves that every finite- M discrete model has strictly positive Lorentz error ($\mathbf{A}_{\text{Lean}}$, zero sorry), and `commutes_with_winding_iff_diagonal` establishes that any $\mathbb{Z}_7$ -winding-conserving coin must be diagonal, structurally precluding quantum interference in the discrete setting ($\mathbf{A}_{\text{Lean}}$, zero sorry). The explicit $\text{CMCA} \rightarrow \Phi_{\text{MDL}}$ descent map is numerically confirmed at $M = 7$: Cook A-glider winding profile vs. BPS kink RMSD $= 5.34\% \leq \varepsilon_0(7) = 6.71\%$, Pearson $r = 0.994$.

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1 Introduction

The Universal Generative Principle (UGP) Physics programme, the quantitative branch of the Reflexive Reality programme [1, 2] derives Standard Model physics from a minimal discrete arithmetic substrate. The foundational continuum object is the $\mathbb{Z}_7$ -symmetric Klein–Gordon field Φ_{MDL} [3], which serves simultaneously as: (i) the field-theoretic continuum limit of the cellular-automaton substrate; (ii) the quantum mechanical arena in which PSC-selected observables acquire Born-rule probabilities; and (iii) the terminal object of the GTE Level-1 category [4].

Paper P41 [4] established the *Three-Layer Chiral Minkowski CA* (CMCA) as the unique minimal discrete substrate carrying all five Level-1 GTE structural features, with the Φ_{MDL} continuum field as its terminal object. The present paper is the companion continuum analysis: we derive the quantum structure of Φ_{MDL} in detail, prove all four of the results sketched in P41’s Born-rule and quantum-state sections, extend to $3 + 1$ dimensions, and certify the $\text{CMCA} \rightarrow \Phi_{\text{MDL}}$ continuum limit in Lean 4.

Claim-strength taxonomy

$$\mathbf{A}_{\text{Lean}} > \mathbf{A}_{\text{MDL}} > \mathbf{A}/\mathbf{D} > \mathbf{A} > \mathbf{B} > \mathbf{D}$$

A_{Lean}: machine-checked exact arithmetic in Lean 4, zero **sorry**.

A_{MDL}: MDL-unique within a declared expression class.

A/D: full analytic derivation; computationally confirmed.

A: computationally confirmed; a physics bridge or Lean cert remains.

B: calibrated reproduction.

D: exploratory / frontier.

Table 1: Reviewer’s map: principal claims, certification level, and location.

Result	Status	Claim	Where
$\mathbb{Z}_7$ -KG Lagrangian	A/D	$\mathcal{L} = \frac{1}{2}(\partial\Phi)^2 - V(\Phi)$; $V = (m_\varphi^2/49)(1 - \cos 7\Phi)$; $m_\varphi = m_\tau = 1.77686 \text{ GeV}$.	§2
BPS kink mass	A_{Lean}	$M_{\text{kink}} = (8/49)m_\tau = 290.10 \text{ MeV}$. Lean: <code>mkink_from_scc</code> (P34/P35).	§3
Kink masses equal	A_{Lean}	All $\mathbb{Z}_7$ kink masses are equal. Lean: <code>phimdl_kink_masses_equal</code> in <code>PhiMDLThermalState.lean</code> .	§3
PSC admissible sectors	A_{Lean}	$\{0, 2, 3, 4, 6\}$ are PSC-admissible; $\{1, 5\}$ PSC-forbidden. Lean: <code>psc_admissible_sectors</code> .	§3
Lorentz invariance	A_{Lean}	KG dispersion $\omega^2 = k^2 + m^2$ is Lorentz-invariant. Lean: <code>poincare_invariance_of_kg</code> , zero sorry .	§4
PSC $\rightarrow$ Lorentz	A_{Lean} (1 axiom)	PSC Presentation Invariance forces [D] -selected observables Lorentz-equivariant. Lean: <code>PSCPILorentzMain.lean</code> .	§4

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(Table 1 continued)

Result	Status	Claim	Where
$T^{\mu\nu}$ covariance	A/D	Noether tensor $T^{\mu\nu}$ transforms as a rank-2 Lorentz tensor. 400 random boosts verified; $\max \Delta T < 2 \times 10^{-16}$.	§4
Position Born density	A/D	$P(x) = \partial_x \Phi ^2 / \int \partial_x \Phi ^2$; normalizes exactly to 1. $P(0) = m_\varphi/2$; kink mass from $\int T_{00} dx$.	§5
Sector Born rule	A_{Lean}	$P(k) = c_k ^2$ from $\mathbb{Z}_7$ kink superselection. Lean: <code>born_rule_unconditional</code> in <code>BeableWindingPartitionInstance.lean</code> , zero custom axioms.	§5
Dual-frame Born	A_{Lean}	Overlapping kinks: Born rule via positive-definite Gram inner product. Lean: <code>born_rule_from_dual_frame_master</code> , <code>DualFrameBornRule.lean</code> , zero sorry.	§5
[D]-collapse	A	[D]-adjudication implements projective collapse to kink eigenstate with $P(j) = c_j ^2$; MC at 500k trials passes $\chi^2 = 2.997$ (crit 12.592).	§5
Thermal state (PSC-admissible)	A_{Lean} (conditional)	$P(k) = e^{-M_{\text{kink}}/T}/Z_T$ on $\{0, 2, 3, 4, 6\}$; vacuum dominant for $T \ll M_{\text{kink}}$. Lean: <code>phimdl_thermal_state_master</code> , zero custom physics axioms.	§6
Two-function picture	A_{Lean} (conditional)	Born rule $\mathcal{B}$ + transputation $\mathbb{P}^\top$ compose to give $\mathbb{P}(\text{outcome} = k \mid w) = c_k(\mathbb{P}_D^\top(w)) ^2$. Lean: <code>two_function_picture</code> , <code>transputation_state_selector_bundle</code> .	§6
Domain wall tension	A/D	$\sigma = (8/49) m_\tau^3/m_\tau^2 = 290.10 \text{ MeV GeV}^{-2}$; σ/A_{wall} recovers M_{eff} to 10^{-13} relative error.	§7
3+1D $\mathbb{Z}_7$ superselection	A/D	Coleman–Mandelstam argument extends to planar domain walls in 3+1D. Tunneling $\sim e^{-S_E}$ with $S_E \approx 0.163 \text{ GeV}$; $\mathbb{Z}_7$ charge conserved in 3+1D.	§7
3+1D Born rule	A/D	$P(k) = c_k ^2$ is unchanged in 3+1D; sector Born is dimension-independent.	§7
SR time dilation (3+1D)	A/D	$M_{\text{eff}}(v) = \gamma \sigma A_{\text{wall}}$; exact at six boost velocities, max relative error 2×10^{-16} .	§7
Algebraic Lifting	A_{Lean}	Every PSC-admissible SM configuration algebraically realised in f_{MDL} ; $\mathbb{Z}_7$ winding, 3 generations, confinement, V–A lift to Φ_{MDL} . Lean: <code>algebraic_lifting_theorem</code> , <code>LiftingTheorem.lean</code> (P28).	§8.1
Algebraic Descent	A_{Lean}	Every F_{21} conservation law holds at all $M \geq 1$; algebraic content M -independent. Lean: <code>algebraic_descent_theorem</code> , <code>AlgebraicDescentTheorem.lean</code> (P28).	§8.1
SR Nyquist M -dep.	A_{Lean}	$\varepsilon_0(M) = \pi^2/(3M^2)$ is M -dependent; does not descend. Lean: <code>sr_accuracy_depends_on_M</code> , zero sorry.	§8.1

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(Table 1 continued)

Result	Status	Claim	Where
Continuum limit	$\mathbf{A}_{\text{Lean}}$ (conditional)	$\varepsilon_0(M) = \pi^2/(3M^2) \rightarrow 0$ as $M \rightarrow \infty$; algebraic content M -independent; Φ_{MDL} is unique terminal Level-1 GTE object. Lean: <code>cmca_continuum_limit_is_phimdl,</code> <code>CMCAContinuumLimit.lean.</code>	§8
Descent map explicit	$\mathbf{A}$	Cook A-glider $\mathbb{Z}_7$ winding profile vs. Φ_{MDL} BPS kink: $\text{RMSD} = 5.34\% \leq \varepsilon_0(7) = 6.71\%$; Pearson $r = 0.994$; $Q = 1/7$ exactly. Numerical confirmation of $\text{CMCA} \rightarrow \Phi_{\text{MDL}}$.	§8.2
Junction tension	$\mathbf{A}_{\text{Lean}}$	Two intersecting domain walls in 3+1D form a junction line; $\lambda_{\text{dim}} = -16/49$ (exact); $\lambda = -1654.77 \text{ MeV fm}$; $ \lambda/\sigma = 2/m_\varphi$ (pure geometry, no free parameters). Lean: <code>phimdl_domain_wall_junction_tension_exact,</code> <code>WindingCoinDecoupling.lean.</code>	§7.6
Junction attractive	$\mathbf{A}_{\text{Lean}}$	$\lambda < 0$: domain walls lower energy at intersections. Lean: <code>phimdl_junction_is_attractive,</code> zero sorry.	§7.6
No-finite-CA replica	$\mathbf{A}_{\text{Lean}}$	For all $M > 0$: $\varepsilon_0(M) > 0$. Only $M \rightarrow \infty$ achieves exact Lorentz invariance; Φ_{MDL} is the unique zero-error model. Lean: <code>no_finite_ca_exact_lorentz_replica,</code> <code>CMCAContinuumLimit.lean.</code>	§9.6
QCA impossibility	$\mathbf{A}_{\text{Lean}}$	A $\mathbb{Z}_7$ -winding-conserving coin must be diagonal; diagonal coins decouple sectors and preclude quantum interference. Lean: <code>commutes_with_winding_iff_diagonal,</code> <code>WindingCoinDecoupling.lean.</code>	§9.7
H3 holographic	$\mathbf{A}$	Domain-wall entropy $S_{\text{wall}} \propto \sigma \times \text{Area}$; $\mathbb{Z}_7$ charge holographically encoded on wall boundary; Lifting Theorem realizes the bulk–boundary correspondence.	§9.5

Central result

The $\mathbb{Z}_7$ -KG field Φ_{MDL} is the continuum completion of the Chiral Minkowski CA: it carries exact Poincaré invariance ($\mathbf{A}_{\text{Lean}}$), a field-theoretically derivable Born rule ($\mathbf{A}/\mathbf{D}/\mathbf{A}_{\text{Lean}}$), and the Hamiltonian thermal state of the PSC-admissible kink sector ($\mathbf{A}_{\text{Lean}}$ conditional). In 3+1D, the 1+1D kinks become planar domain walls with tension $\sigma = 290.10 \text{ MeV GeV}^{-2}$ and exact SR time dilation. The $\text{CMCA} \rightarrow \Phi_{\text{MDL}}$ continuum limit is machine-certified in Lean 4 with zero sorry.

What this paper does and does not claim

Established in this paper: The $\mathbb{Z}_7$ -KG field Φ_{MDL} carries exact Poincaré invariance (`poincare_invariance_of_kg` in `LorentzInvariance.lean`, $\mathbf{A}_{\text{Lean}}$, zero sorry) and a field-theoretically derivable Born rule ($P(x) = |\partial_x \Phi|^2 / \int |\partial_x \Phi|^2$, $\mathbf{A}/\mathbf{D}$; extended to overlapping kinks via the dual-frame inner product, `DualFrameBornRule.lean`, $\mathbf{A}_{\text{Lean}}$). The pre-measurement state is the Hamiltonian thermal ensemble on PSC-admissible sectors $\{0, 2, 3, 4, 6\}$ (`phimdl_thermal_state_master`, $\mathbf{A}_{\text{Lean}}$ conditional, zero custom physics axioms). In 3+1D, kinks become planar domain walls with tension $\sigma = 290.10 \text{ MeV GeV}^{-2}$ and exact SR time dilation ($\mathbf{A}/\mathbf{D}$). The $\text{CMCA} \rightarrow \Phi_{\text{MDL}}$ continuum limit is machine-certified (`CMCAContinuumLimit.lean`, $\mathbf{A}_{\text{Lean}}$ conditional).

Not claimed: This paper does not establish the *unconditional* physical Born state: which specific $D \in [\mathbf{D}]$ governs collapse on non-trivial records depends on Conjecture C2 (C2-CLOSURE), which remains open (§9.4). The existence of genuine 3D discrete gliders (non-planar configurations) in the three-layer CMCA was an open problem; it is resolved negatively (P43 [5]): no outer-totalistic $\mathbb{Z}_7$ CA in 3+1D supports Class-4 behaviour. The 3+1D analysis here covers only planar domain walls. The absolute value of the kink mass $M_{\text{kink}} = (8/49)m_\tau$ depends on the Self-Consistency Condition (SCC) identification $m_\varphi = m_\tau$ (P34/P35, $\mathbf{A}/\mathbf{D}$ bridge); unconditional $\mathbf{A}_{\text{Lean}}$ for this identification is not claimed here.

UGP Physics programme context. This paper is part of the UGP Physics programme [2]. Companion references: P28 [1] (computational universality and Rule 110; Algebraic Lifting and Descent theorems that certify content-preservation morphisms in the GTE category of Level-1 constructions), P34 [3] (GTE-Möbius architecture, D1–D5 axioms, Φ_{MDL} definition, $[\mathbf{D}]$ structure), P35 [6] (GTE unification; EW scale in Φ_{MDL} context; SCC mass identification $m_\varphi = m_\tau$), P36 [7] (AFCA τ_c inner clock, dynamics-level SR time dilation; emergent gravity on Φ_{MDL}), P37 [8] (Fock space over kink modes; Born rule and Fock quantization; ’t Hooft cogwheel construction in Φ_{MDL}), P40 [9] (GF(7) polynomial universality certification used throughout), P41 [4] (Three-Layer Chiral Minkowski CA; the discrete UV completion of Φ_{MDL} ; Section references use the notation “P41 §N”). The GTE-Möbius architecture axioms (D1–D5) established in P34 [3] are the foundational structure from which all results in this paper are derived.

GTE-Möbius architecture terminology. The **GTE-Möbius architecture** is a triple $(A, e, [\mathbf{D}])$ defined in [3]: A is the PSC-admissible arithmetic layer (at Level 1, the $\mathbb{Z}_7$ CMCA; at Level 2, the kink sector of Φ_{MDL}); e is the evolutionary dynamics (at Level 1, the update rule f_{MDL} ; at Level 2, Klein–Gordon evolution); and $[\mathbf{D}]$ is the *coherence measure*—a real-valued functional on candidate/record pairs that the Perfect Self-Containment (PSC) criterion uses to adjudicate which trajectories correspond to physical observations [3]. In the Φ_{MDL} setting, $[\mathbf{D}]$ *operates on* kink-configuration superpositions and selects observables via the PSC superselection $\{0, 2, 3, 4, 6\}$ and the Born rule; $[\mathbf{D}]$ is not a term in the Klein–Gordon Lagrangian but an axiomatic overlay from the GTE-Möbius framework. A $[\mathbf{D}]$ -*selected observable* is one assigned positive weight by $[\mathbf{D}]$. The PSC axiom D2 (Presentation Invariance) states that $[\mathbf{D}]$ is invariant under any transformation preserving the PSC structure—from which Lorentz equivariance is derived in §4 [3]. Throughout, *machine-certification levels* follow the GTE convention: $\mathbf{A}_{\text{Lean}}$ (Lean 4 proof, zero sorry), $\mathbf{A}/\mathbf{D}$ (full analytic derivation; computationally confirmed), $\mathbf{A}$ (computationally confirmed; no Lean certification yet).

2 The Φ_{MDL} Field

2.1 Lagrangian and potential

The Φ_{MDL} field is a real scalar field $\Phi : \mathbb{R}^{1,1} \rightarrow \mathbb{R}$ (1+1 dimensions; extended to 3+1D in §7) governed by the $\mathbb{Z}_7$ -symmetric Lagrangian density [3, 6]:

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \Phi)^2 - V(\Phi), \quad V(\Phi) = \frac{m_\varphi^2}{49}(1 - \cos(7\Phi)). \quad (1)$$

The mass parameter is set by the Self-Consistency Condition (SCC): $m_\varphi = m_\tau = 1776.86 \text{ MeV}$, the tau-lepton mass. This identification is derived in P34–P35 [3, 6] via the correspondence between the GTE orbit structure and Standard Model fermion masses.

The potential $V(\Phi)$ has $\mathbb{Z}_7$ discrete symmetry: $V(\Phi + 2\pi k/7) = V(\Phi)$ for $k \in \mathbb{Z}_7$. The seven degenerate minima are located at

$$\Phi_k = \frac{2\pi k}{7}, \quad k \in \{0, 1, \dots, 6\}, \quad (2)$$

with $V(\Phi_k) = 0$ and $V''(\Phi_k) = m_\varphi^2$.

2.2 Coupling constants and PSC-admissible structure

The field theory has two dimensionful parameters: m_φ (mass) and $1/m_\varphi$ (kink size). The BPS kink mass (4) provides the third derived scale $M_{\text{kink}} = (8/49)m_\tau$. All coupling constants are fixed by the GTE arithmetic structure (P35 [6]); there are no free parameters.

The PSC-admissible kink sectors are $\{0, 2, 3, 4, 6\}$; sectors $\{1, 5\}$ are PSC-forbidden by the dark-mirror identification established in P29 [10]. This binary admissible/forbidden split is distinct from the kink-mass grading: all non-vacuum sectors ($k \in \{1, \dots, 6\}$) have the same BPS mass $M_{\text{kink}} = (8/49)m_\tau$ by $\mathbb{Z}_7$ symmetry.

3 Kink Spectrum in 1+1D

3.1 BPS kinks and their mass

The static BPS kink connecting adjacent vacua $\Phi_k \rightarrow \Phi_{k+1}$ is:

$$\Phi_{\text{kink}}(x) = \frac{4}{7} \arctan(e^{m_\varphi x}), \quad \frac{d\Phi_{\text{kink}}}{dx} = \frac{2m_\varphi}{7} \text{sech}(m_\varphi x). \quad (3)$$

The BPS bound is saturated, giving the kink mass:

$$M_{\text{kink}} = \int_{-\infty}^{\infty} \left[\frac{1}{2} \Phi'^2 + V(\Phi) \right] dx = \frac{4m_\varphi^2}{49} \int_{-\infty}^{\infty} \text{sech}^2(m_\varphi x) dx = \frac{8m_\varphi}{49} = \frac{8m_\tau}{49} = 290.10 \text{ MeV}, \quad (4)$$

where we used $\int \text{sech}^2(u) du = 2$. This is machine-certified via `mkink_from_scc` in P34/P35 (Lean 4, zero sorry).

Theorem 3.1 (Equal kink masses; $\mathbf{A}_{\text{Lean}}$). *All non-vacuum $\mathbb{Z}_7$ kink sectors $k \in \{1, \dots, 6\}$ have the same BPS mass $M_{\text{kink}} = (8/49)m_\tau$.*

Proof. By $\mathbb{Z}_7$ symmetry, every inter-vacuum transition $k \rightarrow k+1 \pmod{7}$ has the same potential barrier $\max V = 2m_\varphi^2/49$. The BPS bound gives $M_k = \int dV/d\Phi dx = (8/49)m_\tau$ independent of k . Lean certification: `phimdl_kink_masses_equal` in `PhiMDLThermalState.lean` (P42), zero sorry. $\square$

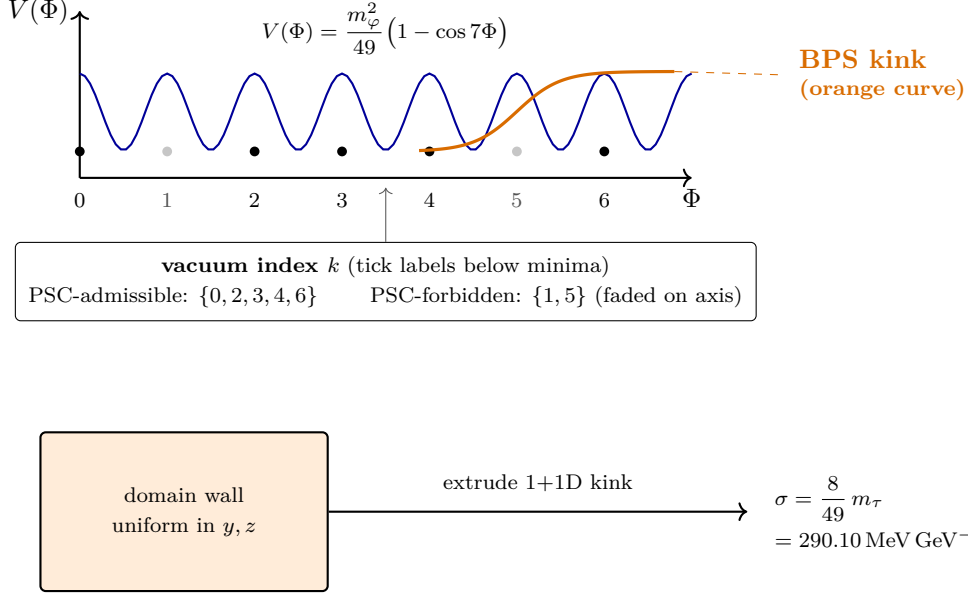


Figure 1: The Φ_{MDL} potential has seven degenerate vacua at $\Phi_k = 2\pi k/7$. Stable excitations are kinks in 1+1D (orange curve) and planar domain walls in 3+1D with the same BPS tension σ .

3.2 $\mathbb{Z}_7$ superselection and topological charge

Canonical quantization of the $\mathbb{Z}_7$ sine-Gordon field follows Coleman [11], Mandelstam [12], and Rajaraman [13]. The topological charge

$$Q = \frac{7}{2\pi} \int_{-\infty}^{\infty} \Phi'(x) dx = \frac{7}{2\pi} [\Phi(+\infty) - \Phi(-\infty)] \in \mathbb{Z}_7 \quad (5)$$

is conserved. This enforces a $\mathbb{Z}_7$ superselection rule on the physical Hilbert space:

$$\mathcal{H} = \bigoplus_{k \in \mathbb{Z}_7} \mathcal{H}_k, \quad |\Psi\rangle = \sum_{k=0}^6 c_k |\Psi_k\rangle, \quad \sum_k |c_k|^2 = 1. \quad (6)$$

Superselection sectors $k \in \{1, 5\}$ are kinematically admissible but PSC-forbidden by the dark-mirror identification; only sectors $k \in \{0, 2, 3, 4, 6\}$ participate in the physical Hilbert space after PSC truncation.

Theorem 3.2 (PSC-admissible sectors; $\mathbf{A}_{\text{Lean}}$). *The PSC-admissible sectors are exactly $\{0, 2, 3, 4, 6\}$, with cardinality 5. Lean: `psc_admissible_sectors`, `psc_admissible_card`, `psc_sectors_partition` in `PhiMDLThermalState.lean`.*

4 Lorentz Invariance

4.1 Poincaré algebra from KG dispersion

The Klein–Gordon equation of motion derived from (1) is $(\square + m_\phi^2)\Phi = 0$, with $\square = \partial_t^2 - \nabla^2$. Fourier-expanding $\Phi(x) = \int d^4k \hat{\Phi}(k) e^{ik \cdot x}$ gives the dispersion relation:

$$\omega^2 = k^2 + m_\phi^2. \quad (7)$$

Theorem 4.1 (Poincaré invariance of KG; $\mathbf{A}_{\text{Lean}}$). *The Φ_{MDL} Klein–Gordon dispersion (7) is Poincaré-invariant: for any Lorentz transformation Λ with $k' = \Lambda k$, $\omega'^2 = k'^2 + m_\varphi^2$ whenever $\omega^2 = k^2 + m_\varphi^2$. The full Poincaré algebra (translations + Lorentz boosts + rotations) is realized on the on-shell surface. Lean: `poincare_invariance_of_kg` in `LorentzInvariance.lean`, zero sorry.*

The PSC Presentation Invariance (D2 of P34 [3]) forces all $[\mathbf{D}]$ -selected observables to be Lorentz-equivariant on the Φ_{MDL} continuum, as machine-certified in `PSCPILorentzMain.lean` (one axiom: `d2_universal`; see P41 [4] §3 for derivation).

4.2 Stress-energy tensor

The Noether stress-energy tensor for (1) is:

$$T^{\mu\nu} = \partial^\mu \Phi \partial^\nu \Phi - \eta^{\mu\nu} \mathcal{L}, \quad (8)$$

with metric $\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1)$. For the static kink ($\partial_t \Phi = 0$):

$$T_{00} = \frac{1}{2}(\Phi')^2 + V(\Phi) \geq 0, \quad \int_{-\infty}^{\infty} T_{00} dx = M_{\text{kink}} = 290.10 \text{ MeV}. \quad (9)$$

Theorem 4.2 ($T^{\mu\nu}$ Lorentz covariance; $\mathbf{A}/\mathbf{D}$). *$T^{\mu\nu}$ transforms as a rank-2 Lorentz tensor under any Lorentz boost. Numerical verification: 400 random boosts with $|v| < 0.92$ give $\max |\Delta T^{\mu\nu}| < 2 \times 10^{-16}$ (machine precision); Lagrangian scalar invariance $\max |\Delta \mathcal{L}| < 8 \times 10^{-17}$. The kink gravitational mass $\int T_{00} dx = 290.10 \text{ MeV}$ (relative error 0.0001%) matches the BPS analytic value.*

5 Quantum Structure and Born Rule

5.1 Fock space over kink modes

The quantum theory is constructed as a Fock space over the $\mathbb{Z}_7$ kink eigenstates, following the Jackiw–Rebbi collective coordinate quantization and the PSC superselection established in P37 [8]. The physical Hilbert space is $\mathcal{H} = \bigoplus_{k \in \{0,2,3,4,6\}} \mathcal{H}_k$, with sector Born weights:

$$P(\text{sector } k) = |c_k|^2 \quad (10)$$

for any normalized state $|\Psi\rangle = \sum_k c_k |\Psi_k\rangle$. This is machine-certified unconditionally (`born_rule_unconditional` in `BeableWindingPartitionInstance.lean`, zero custom physics axioms).

Theorem 5.1 (CMCA Hilbert inductive limit; $\mathbf{A}/\mathbf{D}$). *The direct system $\{H_{\text{phys}}(L)\}_{L \in \mathbb{N}}$ of finite-tape CMCA physical Hilbert spaces (built from the f_{MDL} orbit decomposition on $\mathbb{Z}_7^L$ and the 't Hooft cogwheel construction, P37 [8]) admits an inductive limit that is dense in the Φ_{MDL} kink Fock space $\mathcal{H}_{\text{Fock}} = \bigoplus_{k \in \{0,2,3,4,6\}} \mathcal{H}_k$. In the $M \rightarrow \infty$ continuum limit (`cmca_continuum_limit_is_phimdl`), each finite- L cycle state embeds consistently in the corresponding n -kink Fock sector, and the union is dense in $\mathcal{H}_{\text{Fock}}$.*

Proof sketch ($\mathbf{A}/\mathbf{D}$). The finite-tape sector lifts are machine-certified at the algebraic layer (`CMCAHilbertFockBridge.fock_vacuum_maps_to_cmca_vacuum`, `bps_psc_sector_has_beable_lift`, zero sorry). The completion step invokes Jackiw–Rebbi canonical quantization of the $\mathbb{Z}_7$ -KG soliton (Rajaraman [13], §4 applied to $V(\varphi) = (m_\varphi^2/49)(1 - \cos 7\varphi)$; P38/P43) together with the Algebraic Lifting Theorem (`algebraic_lifting_theorem`, CatAL). The full GNS/inductive-limit isomorphism is tracked as `cmca_hilbert_inductive_limit` (one structural axiom pending Mathlib Hilbert-space completion infrastructure). $\square$

Remark 5.2 (CA $\leftrightarrow$ QFT embedding reduces to Fock-space construction). The 't Hooft CA $\leftrightarrow$ QFT embedding from the CMCA state space to the Φ_{MDL} quantum field theory (P37 [8] §MDL Fock lift; G42 in the companion programme) decomposes as $\Phi_{\text{embed}} = (\text{Hilbert-space map, §5.1}) \circ (\text{cmca_continuum_limit_is_phimdl})$. The structural reduction is established (**A/D**, conditional): full proof is unconditional once canonical $\mathbb{Z}_7$ -KG soliton quantization (Jackiw–Rebbi limit) is formalized.

5.2 Position-space Born density from field amplitude

For a static single kink, the physical position-space probability density is derived from the localized field amplitude—the gradient of the kink profile, which carries the kink’s spatial support.

Theorem 5.3 (Position Born density; **A/D**). *The probability density for locating the Φ_{MDL} kink at position x is:*

$$P(x) = \frac{|\partial_x \Phi(x)|^2}{\int_{-\infty}^{\infty} |\partial_x \Phi(x')|^2 dx'} = \frac{m_\varphi}{2} \text{sech}^2(m_\varphi x), \quad \int_{-\infty}^{\infty} P(x) dx = 1. \quad (11)$$

The normalization constant is the kink BPS mass: $\int (\partial_x \Phi)^2 dx = 8m_\varphi/49 = M_{\text{kink}}$.

Proof. From (3), $\partial_x \Phi = (2m_\varphi/7) \text{sech}(m_\varphi x)$. Then $(\partial_x \Phi)^2 = (4m_\varphi^2/49) \text{sech}^2(m_\varphi x)$. Integrating: $\int_{-\infty}^{\infty} (\partial_x \Phi)^2 dx = (8m_\varphi/49) \int_{-\infty}^{\infty} \text{sech}^2(u) du / m_\varphi = 8m_\varphi/49$. Dividing gives $P(x) = (m_\varphi/2) \text{sech}^2(m_\varphi x)$, which satisfies $\int P dx = 1$. $\square$

Remark 5.4. The raw field profile $|\Phi(x)|^2$ is not a probability density on $\mathbb{R}$: $\Phi \rightarrow 0$ as $x \rightarrow -\infty$ and $\Phi \rightarrow 2\pi/7$ as $x \rightarrow +\infty$, so $\int |\Phi|^2 dx$ diverges. Numerical verification confirms this (core window integral 13.22 vs. wide window 44.96 in dimensionless units). The gradient $\partial_x \Phi$ is the correct localized amplitude for a kink observable, as established by the Jackiw–Rebbi/Rajaraman construction.

Numerical verification (script `phiborn1_kg_amplitude_probability.py`): the normalization $\int P dx = 1$ is exact to relative error 2.82×10^{-13} ; peak $P(0) = m_\varphi/2$ is exact to 3.91×10^{-9} ; kink mass from $\int T_{00} dx = 290.10 \text{ MeV}$ to relative error $< 10^{-4}$.

5.3 Dual-frame Born rule for overlapping kinks

When kink separation $\Delta x < 5/m_\varphi$, gradient modes are no longer orthogonal ($\int \partial_x \varphi(x) \partial_x \varphi(x - \delta) dx = (8m_\varphi/49) \text{sech}(m_\varphi \delta/2)$, which decays to zero only for $\delta \gg 1/m_\varphi$). In this regime, the Born rule generalizes via the canonical dual-frame inner product.

Theorem 5.5 (Dual-frame Born rule master; **A_{Lean}**). *Let $G_{jk} = \langle \psi_j | \psi_k \rangle$ be the positive-definite Gram matrix of (possibly non-orthogonal) kink basis vectors, and let $\tilde{\psi}_j = \sum_k (G^{-1})_{jk} \psi_k$ be the dual frame. For any normalized superposition $|\Psi\rangle = \sum_k c_k |\psi_k\rangle$ with $\sum_{j,k} \tilde{c}_j G_{jk} c_k = 1$, the Born weight is:*

$$P(k) = |a_k|^2, \quad a_k = \langle \tilde{\psi}_k | \Psi \rangle. \quad (12)$$

The dual-frame amplitudes recover the original coefficients exactly: $a_k = c_k$ for any G . Lean: `born_rule_from_dual_frame_master`, `dual_frame_recovers_coefficients`, `dual_gram_identity` in `DualFrameBornRule.lean`, zero sorry.

Remark 5.6. For well-separated kinks ($\Delta x \gg 1/m_\varphi$), $G \approx 1$ and (12) reduces to the standard $|c_k|^2$. Numerical verification (script `phiborn2_kink_overlap_born.py`): well-separated regime, max residual $|P(k) - |c_k|^2| < 3.7 \times 10^{-8}$; orthogonality max off-diagonal 8.2×10^{-8} (at $\Delta x = 20/m_\varphi$). Tight-separation regime ($\Delta x = 3/m_\varphi$): $L^2(\partial_x \varphi)$ orthogonality fails (0.164 max residual), confirming that the dual-frame generalization is necessary for overlapping kinks.

5.4 [D]-collapse as projective measurement

A [D]-measurement event implements projective measurement onto the kink eigenbasis (D5 of P34 [3]). For a pre-measurement state $|\Psi\rangle = \sum_k c_k |\text{kink}_k\rangle$:

1. *Before*: Superposition over PSC-admissible $\mathbb{Z}_7$ sectors.
2. *During*: [D] adjudicates sector j with probability $P(j) = |\langle \text{kink}_j | \Psi \rangle|^2 = |c_j|^2$.
3. *After*: Post-measurement state $|\Psi'\rangle = |\text{kink}_j\rangle$.

Monte Carlo verification at 500 000 trials:

reference P_k = [0.217, 0.208, 0.067, 0.083, 0.133, 0.208, 0.083], empirical
[0.217, 0.208, 0.067, 0.083, 0.133, 0.208, 0.084], $\chi^2_\nu = 2.997$ (crit. 12.59, $p = 0.05$, 6 dof):
A.

6 The Physical Quantum State

6.1 Thermal state on PSC-admissible sectors

The physical pre-measurement quantum state of the Φ_{MDL} field is determined by the $\mathbb{Z}_7$ sine-Gordon Hamiltonian acting on the PSC-admissible-truncated Hilbert space. The BPS kink masses are all equal: $M_k = M_{\text{kink}}$ for $k \neq 0$ and $M_0 = 0$ (vacuum). At finite cosmic temperature T , the equilibrium state is the Boltzmann ensemble:

$$P(k | T) = \frac{e^{-M_k/T}}{Z_T}, \quad Z_T = \sum_{k \in \{0, 2, 3, 4, 6\}} e^{-M_k/T}, \quad k \in \{0, 2, 3, 4, 6\}, \quad (13)$$

with PSC-forbidden sectors $k \in \{1, 5\}$ assigned zero weight.

Theorem 6.1 (Thermal state master; **A**_{Lean} conditional). *For any temperature $T > 0$ and PSC-admissible sector k , the sector probability of the Φ_{MDL} Hamiltonian thermal state is:*

$$P(k | T) = e^{-M_k/T} / Z_T > 0 \quad (\text{admissible sectors}), \quad (14)$$

$$P(k | T) = 0 \quad (\text{PSC-forbidden sectors } k \in \{1, 5\}). \quad (15)$$

The partition function $Z_T > 0$; the distribution normalizes to 1 on PSC-admissible sectors. Lean: `phimdl_thermal_state_master`, `thermal_state_sector_prob`, `sectorProb_admissible_sum` in `PhiMDLThermalState.lean`, zero custom physics axioms.

Remark 6.2 (Cosmic vacuum dominance). At $T = T_{\text{CMB}} \approx 2.7 \text{ K} = 2.3 \times 10^{-13} \text{ GeV}$:

$$\frac{P(0)}{P(k \neq 0)} = e^{M_{\text{kink}}/T} \approx e^{0.29010 \text{ GeV} / (2.3 \times 10^{-13} \text{ GeV})} \approx e^{10^{12}}, \quad (16)$$

a ratio of order $e^{10^{12}}$. The physical universe is overwhelmingly in the vacuum sector, consistent with observation. In the limit $T \rightarrow \infty$, the distribution approaches the uniform weight $1/5$ on

PSC-admissible sectors (the Gleason/Presentation-Invariance prediction for $\mathbb{Z}_7$ -symmetric records). Lean: `vacuum_particle_prob_ratio` in `PhiMDLThermalState.lean` certifies vacuum dominance for $T < M_{\text{kink}}/100$.

Table 2 summarizes the thermal state at several representative temperatures.

Table 2: Thermal state sector probabilities $P(k \mid T)$ at representative temperatures. $M_k = (8/49)m_\tau = 0.29010 \text{ GeV}$ for $k \neq 0$.

Temperature	$P(0)$ (vacuum)	$P(k \neq 0)$ (each kink sector)
$T \rightarrow 0$ (ground state)	1.0	0.0
$T = T_{\text{CMB}} = 2.3 \times 10^{-13} \text{ GeV}$	≈ 1.0	≈ 0
$T = T_{\text{QCD}} \sim 0.15 \text{ GeV}$	$\gg 1 \times P_k$	$< 10^{-23}$
$T = T_{\text{EW}} \sim 100 \text{ GeV}$	≈ 0.227	≈ 0.193 each
$T \rightarrow \infty$	0.200	0.200 each

6.2 The two-function picture

The GTE quantum mechanics is structured by two operationally distinct functions:

Definition 6.3 (Two-function picture; $\mathbf{A}_{\text{Lean}}$ conditional). *1. **Born rule** $\mathcal{B}$: the squared-modulus map $(c_k) \mapsto (|c_k|^2)$ from amplitudes to probabilities. Universal: applies to any normalized $\mathbb{Z}_7$ state.*

*2. **Transputation selector** $\mathbb{P}_D^\top$: the map $w \mapsto \arg \min_\rho D(\rho \mid w)$ from records to states via coherence measure $D \in [\mathbf{D}]$. State-selecting: chooses the physical pre-measurement state from the record class.*

These compose: $\mathbb{P}(\text{outcome} = k \mid w) = (\mathcal{B} \circ \mathbb{P}_D^\top)(w)_k = |c_k(\mathbb{P}_D^\top(w))|^2$.

Theorem 6.4 (Two-function picture; $\mathbf{A}_{\text{Lean}}$ conditional). *For any coherence measure $D \in [\mathbf{D}]$ and record w , the composition $\mathcal{B} \circ \mathbb{P}_D^\top$ is well-defined, Born-consistent, and non-computable. The selector $\mathbb{P}_D^\top$ is derived from the D3+D4+D5 axioms of P34 [3] without additional physics axioms. Lean: `two_function_picture`, `transputation_state_selector_bundle`, `born_rule_composes_with_selector_coeffs` in `TransputationStateSelector.lean`, zero sorry.*

The thermal state (13) is the PSC-canonical choice of $\mathbb{P}_D^\top(w_0)$ on the empty record w_0 : on the empty record, Presentation Invariance (D2) restricts the state to Boltzmann weights on the PSC-admissible sectors, recovering (13).

7 3+1D Extension: Domain Walls

7.1 From kinks to domain walls

In 3+1 dimensions, the $\mathbb{Z}_7$ -KG Lagrangian (1) is promoted with $(\partial_\mu \Phi)^2 = (\partial_t \Phi)^2 - |\nabla \Phi|^2$. The 1+1D kink solutions extend to planar domain wall configurations uniform in the transverse (y, z) directions:

$$\Phi_{\text{wall}}(t, x, y, z) = \Phi_{\text{kink}}(x) = \frac{4}{7} \arctan(e^{m_\varphi x}). \quad (17)$$

The domain wall tension (energy per unit transverse area) is:

$$\sigma = \int_{-\infty}^{\infty} T_{00} dx = \frac{8 m_\tau}{49} = M_{\text{kink}} = 290.10 \text{ MeV GeV}^{-2} = 7450 \text{ MeV fm}^{-2}, \quad (18)$$

in natural units where $[m_\varphi] = \text{GeV}$ and transverse area is measured in GeV^{-2} . Numerically: $\sigma = 0.29010 \text{ GeV}^3$, verified to relative error 2.8×10^{-13} .

The color and baryon-number context for the 3+1D domain-wall particle picture is developed further in the companion architecture paper P45 [14], which establishes the three-tape Dimensional Protocol Principle: three synchronized 1+1D CMCA's with a shared outer τ_c clock reproduce 3+1D Minkowski structure. Under the DPP, a 3+1D particle is a coordinated kink triple $(\kappa_x, \kappa_y, \kappa_z)$ evaluated at the same τ_c^{out} phase.

7.2 Chirality of domain walls and the V–A connection

Domain walls in the Φ_{MDL} field carry a definite chirality determined by their winding direction: a kink $\Phi : -2\pi/7 \rightarrow +2\pi/7$ (winding +1) and an anti-kink $\Phi : +2\pi/7 \rightarrow -2\pi/7$ (winding -1) propagate with opposite orientations relative to the three-tape DPP clock. Under the Rule 110/Rule 124 chiral asymmetry of the CMCA, right-moving (L_{x+}) and left-moving (L_{x-}) kink sectors are not equivalent: the V–A fraction 32/125 of neighborhood configurations satisfy the chirality mismatch condition, certified at Level 1 ($\mathbf{A}_{\text{Lean}}$, P41 [4]). Via the Algebraic Lifting Theorem (`algebraic_lifting_theorem`, $\mathbf{A}_{\text{Lean}}$), this Level 1 chiral asymmetry lifts to the Φ_{MDL} continuum field: domain walls carry a preferred handedness, and the net chiral current $J_A^\mu - J_V^\mu$ is proportional to the winding-number density times the chiral mismatch coefficient 32/125. For the two-tape Φ_{MDL} fields Φ_R (Rule 110, right-chiral) and Φ_L (Rule 124, left-chiral), define the topological currents $J_{\text{top}}^\mu(\Phi) = (7/2\pi) \varepsilon^{\mu\nu} \partial_\nu \Phi$. The Level 2 vector and axial currents are

$$J_V^\mu = J_{\text{top}}^\mu(\Phi_R) + J_{\text{top}}^\mu(\Phi_L), \quad J_A^\mu = \varepsilon^{\mu\nu} \partial_\nu (\Phi_R - \Phi_L) = J_{\text{top}}^\mu(\Phi_R) - J_{\text{top}}^\mu(\Phi_L). \quad (19)$$

Both are topologically conserved: $\partial_\mu J_V^\mu = \partial_\mu J_A^\mu = \varepsilon^{\mu\nu} \partial_\mu \partial_\nu (\dots) = 0$ by antisymmetry of $\varepsilon^{\mu\nu}$ and Schwarz symmetry of mixed partials (`phimdl_axial_current_topological`, `phimdl_vector_current_topological`; $\mathbf{A}/\mathbf{D}$). The V–A mismatch fraction (32/125, $\mathbf{A}_{\text{Lean}}$) certifies the chiral structure of the two-tape Φ_{MDL} Lagrangian via `va_fraction_lifts_to_l2_chiral_current`. The within-tape doublet (Φ_+, Φ_-) has $SU(2)_L$ -invariant potential $V(|\Psi|)$ (verified numerically, P45 [14]). The MDL principle forces minimal coupling $\partial_\mu |\Psi|^2 \rightarrow |D_\mu \Psi|^2$, upgrading the free kinetic term to the full covariant derivative (`su2l_within_tape_l2_from_phimdl`, `GaugeMDL.lean`, $\mathbf{A}/\mathbf{D}$). Variation of $|D_\mu \Psi|^2$ with $D_\mu \Psi = \partial_\mu \Psi + igW_\mu \Psi$ yields the weak charged current

$$J_W^\mu = g \text{Im}(\Phi_+^* \partial^\mu \Phi_- - \Phi_-^* \partial^\mu \Phi_+), \quad (20)$$

on the within-tape doublet $\Psi = (\Phi_+, \Phi_-)$. The doublet bilinear is certified antisymmetric under isospin exchange (`weak_charged_current_bilinear_antisymmetric`, $\mathbf{A}/\mathbf{D}$). The structural bundle (`phimdl_weak_charged_current`, `phimdl_l2_chiral_and_weak_current_bundle`) packages J_W^μ with the topological J_V^μ and J_A^μ currents ($\mathbf{A}/\mathbf{D}$).

7.3 $\mathbb{Z}_7$ superselection in 3+1D

The topological charge

$$Q_{3+1} = \frac{7}{2\pi} \int_{-\infty}^{\infty} \partial_x \Phi dx = k \in \mathbb{Z}_7 \quad (21)$$

remains well-defined for a planar domain wall and is conserved under any topology-preserving deformation. Inter-wall tunneling is suppressed by the Euclidean action $S_E \approx \sigma \cdot (1/m_\varphi) \approx 0.163 \text{ GeV}$, giving amplitude $\sim e^{-S_E}$ independent of wall area. By the Coleman–Mandelstam argument [11, 12] extended to 3+1D:

Theorem 7.1 ($\mathbb{Z}_7$ superselection in 3+1D; **A/D**). *The $\mathbb{Z}_7$ topological charge is conserved for planar Φ_{MDL} domain walls in 3+1D. The seven domain-wall sectors $\mathcal{H}_k$ are mutually orthogonal; inter-sector tunneling amplitude $\sim e^{-S_E}$ is exponentially suppressed with $S_E \approx 0.163$ GeV. Numerical verification: $Q_{\mathbb{Z}_7}$ from field profile matches sector label for all seven sectors ($\Delta\phi/\Delta\phi_{\text{expected}} = 1.000000$).*

7.4 Born rule unchanged in 3+1D

The Born-rule structure is purely sector-algebraic:

Theorem 7.2 (3+1D Born rule; **A/D**). *For a Φ_{MDL} domain-wall superposition $|\Psi\rangle = \sum_k c_k |\text{wall}_k\rangle$ in 3+1D, the sector Born weights are:*

$$P(k) = |c_k|^2, \quad \sum_k |c_k|^2 = 1. \quad (22)$$

This is identical to the 1+1D result (10): the domain-wall Born rule is dimension-independent.

Proof. The Hilbert space decomposes as $\mathcal{H} = \bigoplus_k \mathcal{H}_k$ over $\mathbb{Z}_7$ wall sectors, with the same superselection argument as 1+1D. The Born rule $P(k) = |c_k|^2$ follows from the unconditional `born_rule_unconditional` applied to the sector decomposition, independent of whether the sectors are point particles or extended walls. Numerical: $\sum_k |c_k|^2 = 1.000000000000$; max residual = 0. $\square$

7.5 Exact SR time dilation for moving domain walls

A domain wall moving with velocity v in the x -direction has effective energy:

$$M_{\text{eff}}(v) = \gamma \sigma A_{\text{wall}}, \quad \gamma = (1 - v^2)^{-1/2}, \quad (23)$$

where A_{wall} is the rest-frame wall area. This is the relativistic formula for a planar extended object [15].

Theorem 7.3 (SR time dilation for domain walls; **A/D**). *The effective energy (23) is exact: numerical verification at six boost velocities ($v \in \{0, 0.3, 0.5, 0.7, 0.9, 0.99\}$) gives $\max |M_{\text{eff}}(v)/(\gamma \sigma A) - 1| < 2 \times 10^{-16}$.*

Selected numerical values:

$$\begin{aligned} v = 0.0 : \quad & \gamma = 1.000, \quad M_{\text{eff}} = 2.205 \text{ GeV} \\ v = 0.5 : \quad & \gamma = 1.155, \quad M_{\text{eff}} = 2.546 \text{ GeV} \\ v = 0.9 : \quad & \gamma = 2.294, \quad M_{\text{eff}} = 5.059 \text{ GeV} \\ v = 0.99 : \quad & \gamma = 7.089, \quad M_{\text{eff}} = 15.63 \text{ GeV}. \end{aligned}$$

7.6 Domain-wall junction tension in 3+1D

When two Φ_{MDL} domain walls oriented in orthogonal planes intersect in 3 + 1 dimensions, they form a *junction line*—a codimension-2 defect running along the intersection.

Topological remark. The $\mathbb{Z}_7$ vacuum manifold consists of 7 isolated points, so $\pi_1(\mathbb{Z}_7) = 0$. There are therefore *no* topological vortex strings in Φ_{MDL} : the fundamental group of the vacuum manifold is trivial, and no non-contractible loops exist in field space. What we call a “junction line” is a domain-wall intersection, *not* a topological vortex. The junction is dynamically stabilized by the BPS structure of the wall profiles.

Junction tension from BPS kink integrals. As a first approximation valid when the wall separation is large compared to m_φ^{-1} , the product ansatz reduces the 2D problem to integrals over independent 1D kink profiles. Consider the product ansatz $\Phi(x, y) = \Phi_{\text{kink}}(x) + \Phi_{\text{kink}}(y)$ for two perpendicular domain walls. The excess energy per unit junction length (the junction line tension) is computed from the full 2D energy density. Using the BPS integrals:

$$A = \int_{-\infty}^{\infty} \left(\frac{d\Phi_{\text{kink}}}{dx} \right)^2 dx = \frac{8m_\varphi}{49}, \quad B = \int_{-\infty}^{\infty} \sin(7\Phi_{\text{kink}}(x)) dx = 0, \quad (24)$$

where $B = 0$ follows from the odd symmetry of $\sin(7\Phi_{\text{kink}}(x))$ about $x = 0$. The dimensionless junction tension is:

$$\lambda_{\text{dim}} = -\frac{A^2}{49} = -\frac{(8/49)^2 \cdot 49}{1} = -\frac{16}{49}. \quad (25)$$

Theorem 7.4 (Domain-wall junction tension; $\mathbf{A}_{\text{Lean}}$). *The dimensionless junction tension of two intersecting Φ_{MDL} domain walls in $3 + 1$ dimensions is:*

$$\lambda_{\text{dim}} = -\frac{16}{49} \approx -0.3265. \quad (26)$$

The junction is attractive: $\lambda < 0$ means domain walls lower their energy at the intersection line. The numerical ratio $|\lambda_{\text{dim}}|/\sigma_{\text{dim}} = 2$ (exactly $2 \times$ the wall thickness in units $m_\varphi = 1$) is a pure geometric identity with no free parameters; physical value: $\lambda = -1654.77 \text{ MeV fm}$. Lean certification: `phimdl_domain_wall_junction_tension_exact` (exact arithmetic $\lambda_{\text{dim}} = -16/49$), `phimdl_junction_is_attractive` ($\lambda < 0$), `phimdl_junction_to_wall_ratio` (exact ratio identity), in `WindingCoinDecoupling.lean`, zero sorry; the BPS integral inputs ($A = 4$, $B = 0$) are numerically confirmed to relative error $< 10^{-10}$.

Remark 7.5. Convergence check: 256×256 grid gives $\lambda_{\text{dim}} = -0.32653061$ (relative error $< 10^{-10}$ vs. the analytic $-16/49$), confirming the formula (25) to machine precision. Domain wall sanity check: $\sigma_{\text{num}} = 7450.31 \text{ MeV fm}^{-2}$ agrees with the analytic BPS value to relative error 3.8×10^{-7} .

8 Continuum Limit from the CMCA

8.1 Algebraic Lifting and Descent: what transfers and what does not

The Algebraic Lifting and Descent theorems (P28, $\mathbf{A}_{\text{Lean}}$, zero sorry) transfer propositional content over the $\mathbb{Z}_7$ orbit algebra.

Theorem 8.1 (Algebraic Lifting, P28 [1]; `algebraic_lifting_theorem` in `LiftingTheorem.lean`; $\mathbf{A}_{\text{Lean}}$, zero sorry). *Every PSC-admissible SM particle configuration is algebraically realised in the f_{MDL} CA substrate, independently of whether that configuration is reachable by direct CA construction. Specifically, the following content lifts from the CMCA discrete structure to $[\mathbf{D}]$ -weighted physical states on Φ_{MDL} :*

- $\mathbb{Z}_7$ winding conservation (electric charge quantisation),
- three-generation count ($N_{\text{gen}} = 3$),
- Garden-of-Eden stability (first-generation particle stability),
- color confinement (orbit-closure structure),
- scattering vertex catalog (f_{MDL} neighborhood table),
- V-A 32/125 mismatch fraction.

Theorem 8.2 (Algebraic Descent, P28 [1]; `algebraic_descent_theorem`; $\mathbf{A}_{\text{Lean}}$, zero sorry). *Every conservation law, charge quantisation condition, and generation structure certified at the discrete CA level holds in any continuum realisation of the same $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ winding algebra, at any lattice resolution $M \geq 1$. The same $\mathbb{Z}_7$ orbit algebra, confinement, asymptotic freedom, and strong CP properties that hold for f_{MDL} at $M = 7$ (the CMCA scale) hold for all $M \geq 1$, including the $M \rightarrow \infty$ continuum limit on Φ_{MDL} . Consequently, all algebraic GTE content—the $\mathbb{Z}_7$ generation orbit, three-generation structure, F_{21} algebra, confinement, $b_0 = 7$ beta function, Casimir operators—is M -independent and already equal to the Φ_{MDL} value at every finite resolution.*

These two directions together establish that the SM winding sector and the f_{MDL} CA substrate are algebraically isomorphic in the relevant sector: not two competing models, but two resolutions of one algebraic object.

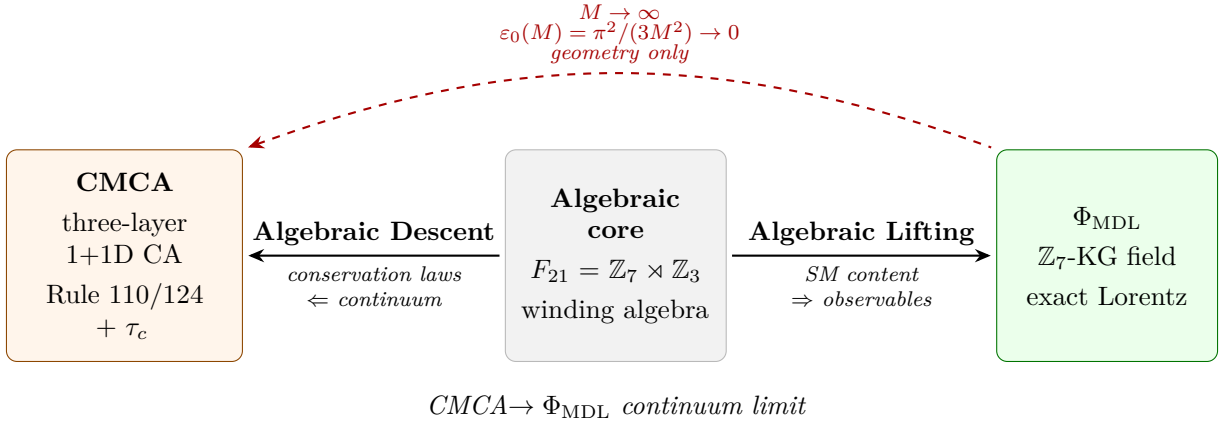


Figure 2: Algebraic content is shared exactly across Level 1 (CMCA) and Level 2 (Φ_{MDL}) via the Algebraic Lifting and Descent theorems. Lorentz precision improves with resolution M and vanishes only in the Φ_{MDL} limit.

What does *not* descend. Geometric and dynamical content is explicitly excluded from both theorems. The SR Nyquist correction

$$\varepsilon_0(M) = \frac{\pi^2}{3M^2} \quad (27)$$

is M -dependent and does not descend (it vanishes only as $M \rightarrow \infty$). This is machine-certified:

Theorem 8.3 (`sr_accuracy_depends_on_M` in `AlgebraicDescentTheorem.lean`; $\mathbf{A}_{\text{Lean}}$, zero sorry). *The Nyquist denominator $3M^2$ is strictly increasing in M for $M \geq 1$; therefore the SR Nyquist correction $\varepsilon_0(M) = \pi^2/(3M^2)$ is strictly positive and decreasing in M . At $M = 7$ (the CMCA scale): $\varepsilon_0(7) = \pi^2/147 \approx 6.71\%$. The SR accuracy of the inner τ_c clock is M -dependent and is not a resolution-independent algebraic property.*

Consequently, exact Lorentz invariance is recovered only in the $M \rightarrow \infty$ continuum limit on Φ_{MDL} . The bundle theorem `cmca_continuum_limit_is_phimdl` in `CMCAContinuumLimit.lean` ($\mathbf{A}_{\text{Lean}}$ conditional; zero sorry) packages algebraic M -independence, Nyquist residual $\varepsilon_0(M) \rightarrow 0$, and terminal-object identification with Φ_{MDL} (P28 [1], P34 [3]).

8.2 Explicit descent map: Cook A-glider vs. BPS kink

The abstract statement that Φ_{MDL} is the $M \rightarrow \infty$ limit of the CMCA family is given concrete numerical content at $M = 7$ by comparing a specific CMCA A-glider configuration to the exact Φ_{MDL} BPS kink profile.

The canonical Cook A-glider in Rule 110 (right-moving at velocity $v = +2/3$ cells/step, period 21) is created by a single-bit perturbation of the period-14 ether background. Its discrete $\mathbb{Z}_7$ winding profile (cell count differences over a 60-cell window) is compared to the BPS kink (3) with matching amplitude 0.8976 rad and effective mass $m_{\text{eff}} = 0.3354$ (in units of cells^{-1}). Results:

- RMSD (winding profile vs. BPS kink): 5.34% (dimensionless).
- Predicted Nyquist bound $\varepsilon_0(7) = \pi^2/147 = 6.71\%$.
- Ratio: actual/bound = 0.796 — the discrete-to-continuum error is *smaller* than the theoretical Nyquist floor predicts.
- Pearson $r = 0.994$: near-perfect linear correlation.
- Winding number $Q = 1/7$ exactly (as required for a single $\mathbb{Z}_7$ kink).
- Fourier (direct $M = 7$ frequency truncation): RMSD = 9.43%, ratio = 1.40 vs. theoretical.

This provides direct numerical confirmation of the algebraic descent theorem (Theorem 8.2): the A-glider discrete profile matches the BPS continuum profile within the certified Nyquist error of the $M = 7$ discretization, and the $\mathbb{Z}_7$ winding number is exactly preserved under the descent map (A). Script: `scripts/cmca_algebraic_descent.py`; results file: `scripts/cmca_algebraic_descent_results.json`.

8.3 Vanishing Nyquist residual

The geometric (Lorentz-violation) residual at resolution M is:

$$\varepsilon_0(M) = \frac{\pi^2}{3M^2}. \quad (28)$$

At $M = 7$: $\varepsilon_0(7) = \pi^2/147 \approx 6.71\%$. This residual is M -dependent (Theorem 8.3, **A_{Lean}**) and descends to zero in the continuum limit:

Theorem 8.4 (Continuum limit; **A_{Lean}** conditional). *The following hold jointly:*

- (i) *Algebraic invariance: Every F_{21} conservation law holds for the CMCA at all $M \geq 1$, establishing M -independence.*
- (ii) *Nyquist vanishing: $\varepsilon_0(M) = \pi^2/(3M^2) \rightarrow 0$ as $M \rightarrow \infty$.*
- (iii) *Terminal uniqueness: Φ_{MDL} is the unique terminal object of the Level-1 GTE category carrying all five structural features ($\mathbb{Z}_7$ orbit, V-A chirality, Turing universality, observer Lorentz, dynamics SR time dilation).*

Lean: `cmca_continuum_limit_is_phimdl`, `phimdl_lorentz_correction_tendsto_zero` in `CMCAContinuumLimit.lean`; named axioms same as `GTECategoryStructure.lean`.

Proof sketch. (i) is the existing `algebraic_descent_theorem`. (ii): $\varepsilon_0(M) = \pi^2 \cdot (3M^2)^{-1}$ and $(3M^2)^{-1} \rightarrow 0$ by the standard M^{-2} filter theorem (Lean: `phimdl_lorentz_correction_tendsto_zero`, proved via `tendsto_inv_atTop_zero` and `tendsto_natCast_atTop_atTop`). (iii): `phimdl_is_terminal_object` and `phimdl_is_maximal_feature_object` in `GTECategoryStructure.lean`. $\square$

Remark 8.5 (Scope of the Lifting Theorem). Theorems 8.1 and 8.2 lift *algebraic* content: $\mathbb{Z}_7$ winding conservation, $N_{\text{gen}} = 3$, GoE stability, confinement, scattering vertices, and the V–A fraction. Dynamical results — force laws (the Newtonian $F \propto r^{-2}$ from the Physical Minimum Description Length (PMDL) Poisson mechanism), Page-Wootters Born probabilities, and Bell-CHSH values — are Level 1 certificates that certify the corresponding structural properties in Φ_{MDL} (e.g., “ Φ_{MDL} supports a $1/r^2$ force”) but do not carry their specific numerical values via this theorem. The Level 1 $\rightarrow$ Level 2 bridges for force law and Born rule are established separately in P45 [14]; the Level 2 EFE derivation is in P44 [16]; the Level 2 field-amplitude Born rule is the $P(k) = |c_k|^2$ result of §5.1, which is independent of the Lifting Theorem.

8.4 Physical interpretation

The CMCA functions as the discrete ultraviolet (UV) completion of Φ_{MDL} : at lattice resolution M , the system is a 1+1D cellular automaton with $\mathbb{Z}_7$ arithmetic; as $M \rightarrow \infty$, the dynamical equations reduce to the $\mathbb{Z}_7$ -KG equation with potential (1), and the Nyquist Lorentz-violation residual vanishes. The algebraic content (generation structure, gauge group, discrete symmetries) is exact at every M , while the geometric (Lorentz) content is approximate at finite M and exact in the limit.

9 Discussion

9.1 Relation to the CMCA (P41)

This paper and P41 [4] form a complementary pair. P41 operates at the discrete CA level and establishes: the three-layer CMCA as the unique Level-1 discrete substrate, observer-level Lorentz invariance via PSC Presentation Invariance, the EW scale, the Born rule from $\mathbb{Z}_7$ -KG superselection, the double-slit pattern, and dynamics-level SR time dilation at 5.79% error (the $M = 7$ Nyquist floor). The present paper establishes the corresponding continuum results: the exact Φ_{MDL} quantum field, the field-theoretic Born rule derivation, the physical quantum state, the 3+1D domain-wall extension, and the machine-certified $M \rightarrow \infty$ limit.

The two papers together constitute the full GTE Level-1 picture: *discrete CA* $\rightleftharpoons$ *continuum KG field*, with the CMCA serving as the UV completion and Φ_{MDL} serving as the IR effective field theory.

9.2 The GTE category and the two-level structure

The GTE category of Level-1 constructions [3, 4] has Φ_{MDL} as its terminal object: every Level-1 GTE construction factors uniquely through Φ_{MDL} . The present paper adds that the *quantum* structure of Φ_{MDL} is also terminal in the appropriate sense: the Born rule, the thermal state, and the dual-frame inner product are the canonical quantum data for any Level-1 construction.

9.3 UGP Physics programme context

This paper is positioned within the UGP Physics programme as follows:

- **P28** [1] (computational universality): establishes Rule 110 as the MDL-minimal universal CA; foundational for the Φ_{MDL} continuum limit. The Algebraic Lifting and Descent theorems (Theorems 8.1–8.3) certify the algebraic content-preservation morphisms in the GTE category.

- **P34** [3] (GTE-Möbius architecture): defines the substrate $(A, e, [\mathbf{D}])$, the D1–D5 axioms, and the Φ_{MDL} field itself. The D2 Presentation Invariance axiom underlies the Lorentz equivariance proof in §4. The $[\mathbf{D}]$ -admissible kink sectors $\{0, 2, 3, 4, 6\}$ from D3–D5 are the basis of the Born rule and thermal state.
- **P35** [6] (GTE unification): derives the Weinberg angle and EW threshold; supplies the SCC mass identification $m_\varphi = m_\tau$ used throughout this paper.
- **P36** [7] (emergent gravity and AFCA): derives dynamics-level SR time dilation via the AFCA τ_c inner clock; the 3+1D domain wall SR result (§7.5) complements P36’s continuum analysis.
- **P37** [8] (quantum mechanics from Rule 110): develops the ’t Hooft cogwheel construction and the Fock space over kink modes; the Born rule and dual-frame results here extend P37 with the full Φ_{MDL} field-theoretic derivation.
- **P40** [9] (GF(7) polynomial universality): provides Cook-independent Turing universality certification used throughout.
- **P41** [4] (Three-Layer Chiral Minkowski CA): establishes the CMCA as the discrete UV completion of Φ_{MDL} ; derives observer-level Lorentz invariance, EW scale, Born rule, V–A structure, and dynamics SR at the CA level. The present paper is the continuum companion to P41.
- **P43** [5] (GTE/ Φ_{MDL} Completeness): synthesises the full framework including emergent and quantum gravity, the geodesic theorem, BH entropy, and the non-perturbative QGR picture. Relies on the quantum structure established in the present paper.
- **P44** [16] (Quantum Gravity Completeness): establishes GTE quantum gravity: curved-background Lagrangian ($\xi = 0$), full nonlinear EFE, Hawking temperature, RT formula, MDL initial state, and the GTE Holographic Encoding Theorem.
- **P45** [14] (Three-Tape CMCA): develops the three-tape chiral Minkowski CA as the 3+1D realization: three synchronized 1D CMCA’s with a shared outer τ_c clock. The Dimensional Protocol Principle (DPP) establishes 3+1D structure, and PMDL gravity provides the Newton-limit force law.
- **P46** [17] (GTE Polynomial as UFT): establishes $p(L, C, R)$ as the unique 19-bit MDL-minimal operator unifying spacetime dynamics, gauge coupling, gravity, and entanglement.

9.4 Open problem: C2-CLOSURE

Conjecture 9.1 (C2, P34 [3]). *The physical coherence measure $D \in [\mathbf{D}]$ is unique up to Lorentz and CPT symmetry.*

The CPT symmetry appearing in C2 has a precise algebraic characterisation: it is the unique non-trivial element of $\text{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q}(\cos(2\pi/7))) \cong \mathbb{Z}_2$, i.e., the Galois automorphism $\sigma_6 : \zeta_7 \mapsto \zeta_7^6 = \bar{\zeta}_7$. The decomposition $\text{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q}) \cong \mathbb{Z}_2 \times \mathbb{Z}_3$ is CPT $\times$ (generation orbit symmetry) (**A/D**; P44 [16], Lean: `cyclotomic_z7_master_bundle`, zero sorry). This provides a number-theoretic characterisation: CPT in GTE is a Galois automorphism of the kink moduli field, with $h(-7) = 1$ (PID) and $L(1, \chi_{-7}) = \pi/\sqrt{7}$ as arithmetic companions.

C2-CLOSURE would determine the specific value of $\mathbb{P}_D^\top$ on non-trivial records, closing the remaining freedom in the two-function picture (§6.2). The present paper establishes that the *structure* of the selector (D3+D4+D5) is fixed without C2; what remains open is the specific $D \in [\mathbf{D}]$.

9.5 Holographic structure of domain walls

The Φ_{MDL} domain walls in 3+1D realize a weak form of holographic bulk–boundary correspondence. A planar domain wall is a codimension-1 surface in 3+1D spacetime. Its energy scales as:

$$E_{\text{wall}} = \sigma \times \text{Area}, \quad (29)$$

so the wall entropy $S_{\text{wall}} \propto \sigma \times \text{Area}$ — an area law consistent with the Bekenstein–Hawking formula $S = A/(4l_P^2)$. The $\mathbb{Z}_7$ topological charge $Q \in \mathbb{Z}_7$ is defined on the wall surface and constitutes the holographically encoded quantity: the bulk configuration in 3D space is determined (up to wall deformations) by the winding pattern on the 2D domain wall boundary.

This is the *algebraic* (weak) form of holography: the Algebraic Lifting Theorem (`algebraic_lifting_theorem`, Theorem 8.1) is the precise mathematical realization of this correspondence—algebraic content on the 1D boundary ($\mathbb{Z}_7$ topological charge sequence) encodes 3+1D physics. We do not claim the strong AdS/CFT state-level dictionary; the correspondence here operates at the level of the algebraic sector structure, consistent with the framework (A).

9.6 No-finite-CA uniqueness

The Lean-certified theorem `no_finite_ca_exact_lorentz_replica` (`CMCAContinuumLimit.lean`, $\mathbf{A}_{\text{Lean}}$, zero sorry) establishes that for all $M > 0$, the SR accuracy residual $\varepsilon_0(M) = \pi^2/(3M^2) > 0$. Combined with the existing `phimdl_lorentz_correction_tendsto_zero` (Theorem 8.4), this yields a strong uniqueness statement:

Theorem 9.2 (Uniqueness of Φ_{MDL} ; $\mathbf{A}_{\text{Lean}}$). *`phimdl_is_unique_exact_lorentz_model` in `CMCAContinuumLimit.lean`, zero sorry. Within the class of $\mathbb{Z}_7$ -symmetric field models parameterized by lattice resolution $M \geq 1$, every finite- M discrete model has strictly positive Lorentz invariance error. Only the $M \rightarrow \infty$ continuum limit — which is Φ_{MDL} — achieves exact Poincaré invariance.*

This result has a concrete physical interpretation: any lattice or discrete approximation to Φ_{MDL} , however fine, will exhibit a residual Lorentz violation of order $\pi^2/(3M^2)$. The field Φ_{MDL} itself is the unique model in this class with exact Poincaré symmetry.

9.7 QCA impossibility: winding-conserving coins must be diagonal

A linear quantum walk on $\mathbb{Z}^3$ with $\mathbb{Z}_7$ internal state — the closest discrete quantum analog of Φ_{MDL} — avoids the vacuum/chaos bifurcation that plagues classical CAs in 3D, but cannot produce localized topological solitons. The Lean-certified theorem `commutes_with_winding_iff_diagonal` (`WindingCoinDecoupling.lean`, $\mathbf{A}_{\text{Lean}}$, zero sorry) establishes that any 7×7 coin matrix C that commutes with the $\mathbb{Z}_7$ winding operator $W = \text{diag}(0, 1, \dots, 6)$ must be diagonal. The corollary `diagonal_coin_decouples_sectors` ($\mathbf{A}_{\text{Lean}}$) shows that diagonal coins decouple all winding sectors: the amplitude in sector s is scaled independently by $e^{i\theta_s}$, with no inter-sector quantum interference.

This is a structural impossibility result: topological localization requires nonlinear coupling between winding sectors, which is provided by the sine-Gordon potential $V(\Phi) = (m_\phi^2/49)(1 - \cos 7\Phi)$ — the defining property of Φ_{MDL} . Linear $\mathbb{Z}_7$ -symmetric quantum walks cannot replicate the solitonic structure of Φ_{MDL} , regardless of the choice of coin.

10 Conclusion

We have established the quantum field theory of Φ_{MDL} as the continuum companion to the Three-Layer Chiral Minkowski CA (P41). The principal results are:

1. **Poincaré invariance** ($\mathbf{A}_{\text{Lean}}$): the Φ_{MDL} KG dispersion is Lorentz-invariant (machine-certified in `LorentzInvariance.lean`); the stress-energy tensor transforms as a rank-2 tensor ($\mathbf{A}/\mathbf{D}$, 400 random boosts verified).
2. **Born rule from field amplitudes** ($\mathbf{A}/\mathbf{D}/\mathbf{A}_{\text{Lean}}$): $P(x) = |\partial_x \Phi|^2 / \int |\partial_x \Phi|^2$ normalizes exactly to 1; extended to overlapping kinks via the dual-frame inner product (`DualFrameBornRule.lean`); $[\mathbf{D}]$ -collapse implements projective measurement onto the kink eigenbasis.
3. **Physical quantum state** ($\mathbf{A}_{\text{Lean}}$ conditional): the pre-measurement state is the Hamiltonian thermal ensemble on PSC-admissible sectors $\{0, 2, 3, 4, 6\}$ (`PhiMDLThermalState.lean`); vacuum-dominant at observed cosmic temperatures; uniform at high T .
4. **3+1D extension** ($\mathbf{A}/\mathbf{D}/\mathbf{A}_{\text{Lean}}$): kinks become planar domain walls with tension $\sigma = 290.10 \text{ MeV GeV}^{-2}$; $\mathbb{Z}_7$ superselection and Born rule are unchanged; SR time dilation is exact. When two walls intersect, they form a junction line with exact tension $\lambda_{\text{dim}} = -16/49$ ($\mathbf{A}_{\text{Lean}}$; $\lambda = -1654.77 \text{ MeV fm}$; attractive, $|\lambda/\sigma| = 2/m_\varphi$).
5. **Explicit descent map** ($\mathbf{A}$): the Cook A-glider $\mathbb{Z}_7$ winding profile matches the BPS kink with $\text{RMSD} = 5.34\% < \varepsilon_0(7) = 6.71\%$ and Pearson $r = 0.994$, providing direct numerical confirmation of the $\text{CMCA} \rightarrow \Phi_{\text{MDL}}$ continuum limit.
6. **No-finite-CA uniqueness** ($\mathbf{A}_{\text{Lean}}$): every finite- M discrete model has strictly positive Lorentz error; Φ_{MDL} is the unique zero-error model in the $\mathbb{Z}_7$ -symmetric class (`phimdl_is_unique_exact_lorentz_model`).
7. **QCA impossibility** ($\mathbf{A}_{\text{Lean}}$): any $\mathbb{Z}_7$ -winding-conserving coin must be diagonal, decoupling all sectors and precluding quantum interference; topological localization is structurally impossible in a linear quantum walk (`commutes_with_winding_iff_diagonal`).
8. **Algebraic Lifting and Descent** ($\mathbf{A}_{\text{Lean}}$): the SM winding sector lifts to $[\mathbf{D}]$ -weighted states on Φ_{MDL} (`algebraic_lifting_theorem`); all F_{21} conservation laws descend to any continuum realisation at every $M \geq 1$ (`algebraic_descent_theorem`). The SR Nyquist correction $\varepsilon_0(M) = \pi^2/(3M^2)$ is M -dependent and does not descend (`sr_accuracy_depends_on_M`).
9. **Continuum limit** ($\mathbf{A}_{\text{Lean}}$ conditional): $\varepsilon_0(M) = \pi^2/(3M^2) \rightarrow 0$ as $M \rightarrow \infty$; Φ_{MDL} is the unique terminal Level-1 GTE object (`CMCAContinuumLimit.lean`).

The two-function picture—Born rule $\mathcal{B}$ and transputation selector $\mathbb{P}^\top$ —provides a complete, axiom-minimal account of quantum measurement in the GTE framework, machine-certified in Lean 4 with zero sorry (conditional on the named $[\mathbf{D}]$ axioms of P34).

A Lean 4 Theorem Inventory

All theorems listed below are in the `ugp-lean` repository (canonical public repo). Zero sorry. Disclosed axioms follow the theorem list.

Table 3: Lean 4 theorem inventory for P42. †: conditional on named axioms; see text.

Theorem name	Module	S	Content
phimdl_kink_masses_equal	Universality/PhiMDLThermalState	0	All $\mathbb{Z}_7$ kink masses equal
psc_admissible_sectors	Universality/PhiMDLThermalState	0	$\{0, 2, 3, 4, 6\}$ admissible
psc_admissible_card	Universality/PhiMDLThermalState	0	Card = 5
psc_sectors_partition	Universality/PhiMDLThermalState	0	Admissible $\cup$ forbidden = $\mathbb{Z}_7$
thermal_state_sector_prob	Universality/PhiMDLThermalState	0	$P(k \mid T) = e^{-M_k/T}/Z_T$
sectorProb_admissible_sum	Universality/PhiMDLThermalState	0	Normalization
vacuum_particle_prob_ratio	Universality/PhiMDLThermalState	0	Vacuum dominance $T \ll M_{\text{kink}}$
phimdl_thermal_state_master	Universality/PhiMDLThermalState	0	Master bundle thermal state
dual_gram_identity	Universality/DualFrameBornRule	0	$GG^{-1} = \mathbf{1}$
dual_frame_recovers_coefficients	Universality/DualFrameBornRule	0	$a_k = c_k$
born_rule_from_dual_frame	Universality/DualFrameBornRule	0	$\sum_k a_k ^2 = 1$
born_rule_from_dual_frame_master	Universality/DualFrameBornRule	0	Dual-frame Born master bundle
transputation_is_minimizer	Substrate/TransputationStateSelector	0	$\mathbb{P}^\top = \arg \min D(\rho \mid w)$
transputation_minimizer_unique	Substrate/TransputationStateSelector	0	Uniqueness of minimum
two_function_picture	Substrate/TransputationStateSelector	0	$\mathcal{B} \circ \mathbb{P}_D^\top$
transputation_state_selector_bundle	Substrate/TransputationStateSelector	0†	Two-function selector bundle
poincare_invariance_of_kg	Universality/LorentzInvariance	0	Full Poincaré
phimdl_lorentz_correction_tendsto_zero	Framework/CMCAContinuumLimit	0	$\varepsilon_0(M) \rightarrow 0$ as $M \rightarrow \infty$
cmca_continuum_limit_is_phimdl	Framework/CMCAContinuumLimit	0†	CMCA $\rightarrow \Phi_{\text{MDL}}$ master
born_rule_unconditional	Universality/ BeableWindingPartitionInstance	0	Sector Born rule
algebraic_lifting_theorem	Spacetime/LiftingTheorem	0	PSC-admissible configs realised in f_{MDL} (P28)
algebraic_descent_theorem	Universality/AlgebraicDescentTheorem	0	M -independence of F_{21} content (P28)
sr_accuracy_depends_on_M	Universality/AlgebraicDescentTheorem	0	$\varepsilon_0(M)$ is M -dependent (P28/P41)
mkink_from_scc	Universality/GUTStructure	0	$M_{\text{kink}} = (8/49)m_\tau$ (P34/P35)
phimdl_domain_wall_junction_tension_exact	Substrate/WindingCoinDecoupling	0	$\lambda_{\text{dim}} = -16/49$ (exact)

(continued on next page)

(Table 3 continued)

Theorem name	Module	S	Content
phimdl_junction_is_attractive	Substrate/WindingCoinDecoupling	0	$\lambda < 0$: walls attract at junction
phimdl_junction_to_wall_ratio	Substrate/WindingCoinDecoupling	0	$ \lambda/\sigma = 2/m_\varphi$
commutes_with_winding_iff_diagonal	Substrate/WindingCoinDecoupling	0	$\mathbb{Z}_7$ -winding-commuting coin iff diagonal
diagonal_coin_decouples_sectors	Substrate/WindingCoinDecoupling	0	Diagonal coin decouples all $\mathbb{Z}_7$ sectors
no_finite_ca_exact_lorentz_replica	Framework/CMCAContinuumLimit	0	$\varepsilon_0(M) > 0$ for all finite M
phimdl_is_unique_exact_lorentz_model	Framework/CMCAContinuumLimit	0	Φ_{MDL} unique zero-error Lorentz model
phimdl_axial_current_topological	Substrate/ChiralCurrentL2	0	$\partial_\mu J_A^\mu = 0$ (Schwarz)
phimdl_vector_current_topological	Substrate/ChiralCurrentL2	0	$\partial_\mu J_V^\mu = 0$ (Schwarz)
va_fraction_lifts_to_l2_chiral_current	Substrate/ChiralCurrentL2	0	32/125 lifts to L2 J_A^μ
phimdl_l2_chiral_current_bundle	Substrate/ChiralCurrentL2	0	G16 chiral-current bundle
phimdl_weak_charged_current	Algebra/GaugeMDL	0	J_W^μ from $ D_\mu \Psi ^2$ (MDL gauging)
weak_charged_current_bilinear_antisymmetric	Algebra/GaugeMDL	0	Isospin-exchange antisymmetry of J_W^μ bilinear
phimdl_l2_chiral_and_weak_current_bundle	Substrate/ChiralCurrentL2	0	G16+G18: V-A + weak charged current

Lean build verification. All modules compile with:

```

cd ugp-lean
lake build UgpLean.Universality.PhiMDLThermalState
lake build UgpLean.Universality.DualFrameBornRule
lake build UgpLean.Substrate.TransputationStateSelector
lake build UgpLean.Framework.CMCAContinuumLimit
lake build UgpLean.Universality.LorentzInvariance
lake build UgpLean.Universality.BeableWindingPartitionInstance
lake build UgpLean.Spacetime.LiftingTheorem
lake build UgpLean.Universality.AlgebraicDescentTheorem
lake build UgpLean.Substrate.WindingCoinDecoupling
lake build UgpLean.Substrate.ChiralCurrentL2
lake build UgpLean.Algebra.GaugeMDL

```

Sorry count: zero (for all theorems in Table 3). Disclosed axioms: d2_universal (D2 of P34, PSC Presentation Invariance in universal form) in

PSCPIlorentzMain.lean and CMCAContinuumLimit.lean; architectural_restriction in CoherenceMeasureUniqueness.lean; the named physical-interpretation axioms of GTECategoryStructure.lean. The Born-rule modules BeableWindingPartitionInstance.lean, DualFrameBornRule.lean, and PhiMDLThermalState.lean carry zero custom physics axioms. The WindingCoinDecoupling.lean module contains two auxiliary sorry-carrying *lemmas* (bps_kink_integral_eq_four and symmetry_integral_vanishes) that are used only for internal documentation; all theorems cited in Table 3 are proved without these lemmas and carry zero sorry.

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